

STAVROPOL, Russian Federation

WMO: 349490

Lat: 45.1084N

Lon: 42.0999E

Elev: 452

StdP: 96.02

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-13.1	-19.8	0.7	-16.9	-15.8	1.0	-12.3	14.5	0.9	12.9	-1.0	4.4	90	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.8	19.4	32.0	19.0	30.1	18.6	21.5	29.2	20.6	28.0	19.8	27.0	5.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.6	24.4	18.1	13.7	23.7	17.2	13.0	23.0	64.4	29.4	61.4	28.2	58.8	26.9	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.8	9.4	-18.3	36.0	5.1	1.8	-22.0	37.3	-25.0	38.4	-27.9	39.4	-31.6	40.7		
(5)			-18.8	22.8	4.9	1.1	-22.3	23.6	-25.2	24.2	-27.9	24.8	-31.5	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-1.9	-1.0	3.8	10.1	15.7	20.4	23.2	23.1	17.5	10.7	4.3	0.1		
	DBStd	10.04	5.70	5.86	4.50	4.24	3.69	3.30	2.86	3.24	3.94	5.00	5.13	5.16		
	HDD10.0	1468	368	308	198	49	3	0	0	0	2	52	181	307		
	HDD18.3	3267	626	541	451	248	96	16	2	3	59	239	420	564		
	CDD10.0	1678	0	1	6	52	181	312	409	406	227	74	11	1		
	CDD18.3	434	0	0	0	1	15	77	152	151	34	3	0	0		
	CDH23.3	4074	0	0	1	10	134	652	1450	1507	291	29	0	0		
	CDH26.7	1531	0	0	1	1	20	209	574	639	79	8	0	0		
(14) Wind	WSAvg	4.5	4.6	4.8	5.2	4.9	4.3	4.1	4.1	4.2	4.4	4.5	4.6	4.7		
(15) Precipitation	PrecAvg	551	29	27	31	46	64	81	56	51	44	45	41	37		
	PrecMax	797	93	74	69	107	131	219	160	167	108	123	86	99		
	PrecMin	378	2	0	2	16	1	1	3	4	7	1	2	4		
	PrecStd	105	19	18	17	23	30	51	36	38	24	30	23	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	14.8	20.2	24.7	28.9	33.2	35.2	35.9	32.2	27.1	20.2	14.2		
		MCWB	6.4	8.4	10.6	13.7	18.0	19.1	20.3	19.4	18.5	15.9	10.8	7.8		
	2%	DB	8.2	11.2	16.1	22.0	26.8	31.1	33.2	34.0	28.9	22.9	16.1	10.9		
		MCWB	4.6	6.7	8.8	13.0	16.9	18.5	19.4	19.4	17.8	14.3	9.6	6.3		
	5%	DB	6.2	8.2	13.1	19.8	24.8	29.0	31.9	32.2	26.6	20.1	13.2	8.2		
		MCWB	3.7	4.8	7.6	12.2	16.2	18.2	19.3	18.8	16.6	13.8	8.8	5.1		
	10%	DB	4.2	6.0	10.8	17.2	22.8	27.1	30.0	30.2	24.2	18.0	11.0	6.2		
		MCWB	2.3	3.5	6.6	10.8	15.6	18.1	18.9	18.5	15.9	13.0	7.9	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	9.1	11.3	15.1	19.4	22.1	22.7	22.4	20.4	16.7	12.2	8.8		
		MCDB	10.3	13.2	18.3	22.0	26.4	28.8	31.3	30.8	27.5	23.1	16.6	13.1		
	2%	WB	5.2	7.0	9.5	13.7	18.0	20.7	21.5	21.3	18.8	15.5	10.8	6.9		
		MCDB	7.4	10.8	14.7	19.9	24.6	27.3	29.6	29.2	25.6	21.0	15.0	9.6		
	5%	WB	3.7	5.1	8.2	12.7	17.0	19.6	20.5	20.5	17.8	14.5	9.5	5.5		
		MCDB	5.6	8.0	12.2	18.2	22.8	26.1	28.1	27.9	23.9	19.0	12.7	7.6		
	10%	WB	2.5	3.5	6.8	11.7	16.1	18.7	19.8	19.7	16.9	13.4	8.1	4.3		
		MCDB	4.1	5.4	9.8	16.6	21.5	25.0	27.1	27.0	22.5	17.2	10.7	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	8.2	9.9	9.8	10.1	11.0	11.6	10.3	8.6	7.0	6.2		
		MCDBR	8.7	10.3	12.9	13.0	12.8	12.9	13.5	14.1	13.4	12.2	11.1	9.1		
	5% WB	MCWBR	6.1	6.6	7.2	6.7	6.3	5.5	5.3	5.5	5.5	6.0	6.6	6.1		
		MCWBR	7.8	9.7	11.4	11.2	11.2	11.0	11.6	11.8	11.3	10.6	9.7	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.356	0.398	0.442	0.409	0.416	0.440	0.439	0.407	0.373	0.338	0.326		
	taud		2.303	2.225	2.158	2.072	2.254	2.275	2.218	2.218	2.277	2.335	2.371	2.316		
	Ebn at Noon		770	809	823	817	861	854	827	812	807	783	750	724		
	Edh at Noon		81	106	130	155	133	131	137	132	115	94	76	73		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.16	3.04	4.63	5.75	6.35	6.39	5.76	4.22	2.60	1.48	1.05		
	RadStd		0.23	0.34	0.29	0.46	0.60	0.37	0.34	0.30	0.43	0.34	0.19	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page